CHAPTER 7

GCD, Bézout's and Gauss's theorems

Euclid's algorithm, coprime integers, Diophantine equations

Objectives. Find a gcd with Euclid's algorithm; obtain a Bézout pair; use Bézout's and Gauss's theorems; find an inverse modulo n; solve an equation $ax + by = c$ in integers.

Prerequisites. Chapter 6 (divisibility, Euclidean division, congruences); programming a loop in Python.

Fluency

Fluency · No calculator, five minutes.

1. Carry out the Euclidean division of 1,053 by 325.
2. Give the greatest common divisor of 84 and 126.
3. Do the numbers 15 and 28 have a common divisor other than 1?
4. Find two integers u and v such that $5u + 3v = 1$.
5. Solve $4x \equiv 0 \pmod 6$ for $x \in \{0, \dots, 5\}$.
6. If 7 divides $5k$, what can be conjectured about k?

Solution · Fluency.

1. $1{,}053 = 3 \times 325 + 78$.
2. 42.
3. No: $15 = 3 \times 5$ and $28 = 2^2 \times 7$.
4. For example $u = 2$ and $v = -3$.
5. $x = 0$ or $x = 3$: $4 \times 3 = 12 \equiv 0 \pmod 6$. One cannot "divide by 4".
6. That 7 divides k: Gauss's theorem will prove it.

Introductory activity

Activity 1 · Measuring one litre with two cans 35 min. We have a 9-litre can, a 7-litre can, both without markings, and a water supply. We may fill a can, empty it, or pour one into the other until one is empty or the other is full.

1. Show that one can get exactly 2 litres in the large can.
2. If we pour in u times 9 litres altogether and throw away v times 7 litres, the quantity left is $9u - 7v$. Find natural numbers u and v such that $9u - 7v = 1$. Deduce how to get 1 litre.
3. The cans are replaced by cans of 6 and 9 litres. Show that every quantity of the form $6u + 9v$, with u and v integers, is a multiple of 3. Can one get 1 litre?
4. With cans of a and b litres, which whole quantities does it seem possible to get? State a conjecture involving the greatest common divisor of a and b.

Solution · Activity.

1. Fill the 9-litre can and pour it into the 7-litre can: 2 litres are left.
2. $9 \times 4 - 7 \times 5 = 36 - 35 = 1$. Fill the large can four times, pouring it each time into the small one, which is emptied whenever it is full (five times in all): 1 litre is left in the large can.
3. $6u + 9v = 3(2u + 3v)$. Every quantity obtained is a multiple of 3: it is impossible to get 1 litre.
4. Conjecture: one obtains exactly the multiples of $\gcd(a, b)$ (within the capacity of the cans). The key tool is the equality $au + bv = \gcd(a, b)$, proved in the course.

Teacher note · Running the activity. Have the students handle the cans (or simulate on paper) before looking for u and v. Question 2 is solved by trial; the extended Euclidean algorithm of the course makes it systematic. Do not try to prove in class that the conjecture is sufficient for the cans: only Bézout's equality is in the syllabus.

Course notes

GCD and Euclid's algorithm

Definition · GCD. Let a and b be two integers, not both zero. The set of their positive common divisors is finite and contains 1; its largest element is called the *greatest common divisor* of a and b, written $\gcd(a,b)$. We say that a and b are *coprime* if $\gcd(a,b)=1$.

Property · Euclid's lemma. Let a and b be two integers with $b \neq 0$. For every integer k, the common divisors of a and b are those of b and $a-kb$. In particular, if r is the remainder of the division of a by $b \geq 1$, then $\gcd(a,b)=\gcd(b,r)$.

Proof. If $d \mid a$ and $d \mid b$, then $d \mid a-kb$. Conversely, if $d \mid b$ and $d \mid a-kb$, then $d \mid (a-kb)+kb=a$. The two sets of common divisors are equal, so their largest elements are too. □

Method · Euclid's algorithm. Replace (a,b) by (b,r) as long as the remainder r is not zero. The remainders form a strictly decreasing sequence of natural numbers, so the algorithm stops. The gcd is the last nonzero remainder. Example:

$$1{,}053 = 3 \times 325 + 78, \qquad 325 = 4 \times 78 + 13, \qquad 78 = 6 \times 13 + 0,$$

so $\gcd(1{,}053, 325) = 13$.

Bézout's theorem

Property · Writing the gcd. Let a and b be two integers, not both zero. There are integers u and v such that $au+bv=\gcd(a,b)$. Moreover, every common divisor of a and b divides $\gcd(a,b)$.

Proof. Let E be the set of positive integers of the form $au+bv$, with u, v integers. E is not empty (it contains a^2+b^2). Let $d=au_0+bv_0$ be its smallest element.

Divide a by d: $a=dq+r$ with $0 \leq r < d$. Then $r=a-q(au_0+bv_0)=a(1-qu_0)+b(-qv_0)$ is of the form $au+bv$. If it were positive, it would belong to E while being smaller than d: impossible. So $r=0$ and $d \mid a$. Likewise $d \mid b$.

Finally every common divisor c of a and b divides $au_0+bv_0=d$, so $c \leq d$. Hence d is the greatest common divisor: $d=\gcd(a,b)$. □

Property · Bézout's theorem. Two integers a and b are coprime if and only if there are integers u and v such that $au+bv=1$.

Proof. If $\gcd(a,b)=1$, the previous theorem provides u and v. Conversely, if $au+bv=1$, every positive common divisor of a and b divides 1, hence equals 1. □

Method · Bézout pair: working back through Euclid's algorithm. With the previous example: $13 = 325 - 4 \times 78 = 325 - 4(1{,}053 - 3 \times 325) = 13 \times 325 - 4 \times 1{,}053$. A suitable pair for $(1{,}053, 325)$ is $(-4, 13)$.

Key point · Extended Euclidean algorithm in Python. The following function returns (d,u,v) with $d=\gcd(a,b)=au+bv$, for natural numbers a and b not both zero.

```
def euclide_etendu(a, b):
    u0, v0, u1, v1 = 1, 0, 0, 1
    while b != 0:
        q, r = a // b, a % b
        a, b = b, r
```

```
        u0, u1 = u1, u0 - q * u1
        v0, v1 = v1, v0 - q * v1
    return a, u0, v0
```

At every step, the two current remainders can be written $au_0 + bv_0$ and $au_1 + bv_1$ in terms of the initial values of a and b.

Gauss's theorem

Property · Gauss's theorem. Let a, b, c be three integers. If $a \mid bc$ and a and b are coprime, then $a \mid c$.

Proof. By Bézout's theorem, there are u and v such that $au + bv = 1$. Multiplying by c: $c = acu + bcv$. Now $a \mid acu$ and $a \mid bc$, so a divides $acu + bcv = c$. □

Property · Corollary. If a and b are coprime and both divide n, then ab divides n.

Proof. $n = ak$. Since $b \mid ak$ and $\gcd(a, b) = 1$, Gauss gives $b \mid k$: $k = bk'$ and $n = abk'$. □

Property · Inverse modulo n. Let $n \geq 2$ and a an integer. There is an integer u such that $au \equiv 1 \pmod{n}$ if and only if $\gcd(a, n) = 1$. Such a u is an *inverse of a modulo n*; it allows us to solve $ax \equiv b \pmod{n}$: $x \equiv ub \pmod{n}$.

Proof. $au \equiv 1 \pmod{n}$ is equivalent to the existence of an integer v such that $au + nv = 1$: this is Bézout's theorem. □

Method · Equation $ax + by = c$ in integers.

1. It has solutions if and only if $\gcd(a, b)$ divides c; we then divide by this gcd.
2. Find a particular solution (x_0, y_0).
3. By subtraction, $a(x - x_0) = -b(y - y_0)$; with a and b coprime, Gauss gives $x - x_0 = bk$, then $y - y_0 = -ak$.
4. Check that the pairs $(x_0 + bk, y_0 - ak)$, $k \in \mathbb{Z}$, are indeed solutions.

Example: $7x + 5y = 1$ has the particular solution $(-2, 3)$; its solutions are the pairs $(-2 + 5k, 3 - 7k)$, $k \in \mathbb{Z}$.

Exercises

Exercise 7.1 · Euclid's algorithm. Find $\gcd(1{,}071, 462)$ with Euclid's algorithm, then simplify the fraction $\dfrac{462}{1{,}071}$.

Solution. $1{,}071 = 2 \times 462 + 147$, $462 = 3 \times 147 + 21$, $147 = 7 \times 21$. The gcd is 21 and $\dfrac{462}{1{,}071} = \dfrac{22}{51}$.

Exercise 7.2 · Bézout pair. Find a pair of integers (u, v) such that $17u + 12v = 1$.

Solution. $17 = 12 + 5$, $12 = 2 \times 5 + 2$, $5 = 2 \times 2 + 1$. Working back: $1 = 5 - 2 \times 2 = 5 - 2(12 - 2 \times 5) = 5 \times 5 - 2 \times 12 = 5(17 - 12) - 2 \times 12 = 5 \times 17 - 7 \times 12$. The pair $(5, -7)$ works: $85 - 84 = 1$.

Exercise 7.3 · Coprime integers. Let n be a natural number. Prove that n and $n + 1$ are coprime, then that $2n + 1$ and $3n + 1$ are coprime too.

Solution. $(n + 1) \times 1 + n \times (-1) = 1$. Next $3(2n + 1) - 2(3n + 1) = 1$. Bézout's theorem concludes in both cases.

Exercise 7.4 · A homogeneous equation. Find all pairs of integers (x, y) such that $15x = 21y$.

Solution. Dividing by 3: $5x = 7y$. So $7 \mid 5x$ and $\gcd(7, 5) = 1$: Gauss gives $x = 7k$, $k \in \mathbb{Z}$. Then $5 \times 7k = 7y$, that is $y = 5k$. Conversely $15 \times 7k = 105k = 21 \times 5k$. The solutions are the pairs $(7k, 5k)$, $k \in \mathbb{Z}$.

Exercise 7.5 · Exercise books and pens. A bookshop sells exercise books at 1,100 ariary and pens at 700 ariary. A customer spends exactly 10,000 ariary. How many exercise books and pens did they buy?

Solution. We look for natural numbers x and y such that $1{,}100x + 700y = 10{,}000$, that is $11x + 7y = 100$. Modulo 7: $4x \equiv 2 \pmod 7$; since $4 \times 2 = 8 \equiv 1$, $x \equiv 4 \pmod 7$. The particular solution $(4, 8)$ works ($44 + 56 = 100$). The integer solutions are $(4 + 7k, 8 - 11k)$, $k \in \mathbb{Z}$. For $x \geq 0$ and $y \geq 0$, we need $k \geq -\frac{4}{7}$ and $k \leq \frac{8}{11}$, so $k = 0$: 4 exercise books and 8 pens.

Exercise 7.6 · Inverse modulo 26.

1. Explain why 7 has an inverse modulo 26 and check that 15 is one.
2. Solve $7x + 3 \equiv 10 \pmod{26}$.

Solution.

1. $\gcd(7, 26) = 1$; $7 \times 15 = 105 = 4 \times 26 + 1$.
2. $7x \equiv 7 \pmod{26}$; multiplying by 15: $x \equiv 105 \equiv 1 \pmod{26}$. The solutions are the integers $1 + 26k$.

Exercise 7.7 · A gcd that depends on n. Let n be a natural number and $d = \gcd(n+5, 2n+3)$.

1. Prove that d divides 7.
2. Find the integers n for which $d = 7$.

Solution.

1. d divides $2(n+5) - (2n+3) = 7$.
2. $d \in \{1, 7\}$. If $d = 7$, then $7 \mid n+5$, that is $n \equiv 2 \pmod 7$. Conversely, if $n \equiv 2 \pmod 7$, then $n+5 \equiv 0$ and $2n+3 \equiv 7 \equiv 0 \pmod 7$, so $d = 7$. Example: $n = 2$ gives $\gcd(7, 7) = 7$.

Problem

Problem · The affine cipher. The letters A to Z are numbered from 0 to 25. To encrypt a letter with number x, compute the remainder y of $7x+3$ in the division by 26, and replace the letter by the letter with number y.

1. Encrypt the word EXPERT.
2. Prove that $y \equiv 7x+3 \pmod{26}$ is equivalent to $x \equiv 15y+7 \pmod{26}$.
3. Decrypt the message TDNZZ.
4. Replace 7 by a in the key. Prove that if $\gcd(a,26) \neq 1$, two different letters may be encrypted as the same letter. How many values of $a \in \{1,\dots,25\}$ give a cipher that can be decrypted?

Solution · Problem.

1. E$=4 \mapsto 31 \equiv 5$ (F); X$=23 \mapsto 164 \equiv 8$ (I); P$=15 \mapsto 108 \equiv 4$ (E); E $\mapsto$ F; R$=17 \mapsto 122 \equiv 18$ (S); T$=19 \mapsto 136 \equiv 6$ (G). The encrypted word is FIEFSG.
2. If $y \equiv 7x+3$, multiplying $y-3 \equiv 7x$ by 15: $x \equiv 15y-45 \equiv 15y+7 \pmod{26}$. Conversely, if $x \equiv 15y+7$, then $7x+3 \equiv 105y+52 \equiv y \pmod{26}$, since $105 \equiv 1$ and $52 \equiv 0$.
3. T$=19 \mapsto 15\times 19+7 = 292 \equiv 6$ (G); D$=3 \mapsto 52 \equiv 0$ (A); N$=13 \mapsto 202 \equiv 20$ (U); Z$=25 \mapsto 382 \equiv 18$ (S). The message is GAUSS.
4. If $d=\gcd(a,26)>1$, let $x'=x+\frac{26}{d}$. Then $ax'=ax+26\times\frac{a}{d} \equiv ax \pmod{26}$: the letters x and $x' \neq x$ (for x and x' in $\{0,\dots,25\}$, for example $x=0$) have the same encryption. If $\gcd(a,26)=1$, the inverse of a modulo 26 allows decryption as in 2. The $a \in \{1,\dots,25\}$ coprime with $26=2\times 13$ are the odd numbers other than 13: there are 12 of them.

Teacher note · Status of the problem. Ciphers (affine, Vigenère, Hill, RSA) are "possible problems" of the syllabus. The Hill cipher is taken up in chapter 9 (matrices), RSA in chapter 8. Have the cipher programmed in Python.